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Every maths trick — 92 passages, full lesson version

Each passage is now a mini-lesson: a challenge to hook them in, the trick, one worked example, two for the child to try, then both checked together — with encouragement all the way through. Read it like you're at the kitchen table and someone just asked "go on then, show me".

The four rules

1. Before each passage say its number clearly — "Number one", "Number two" — pause a second, read the passage, then pause a couple of seconds before the next number. Those gaps are how I find the joins. 2. Wherever you see ... leave a short beat — two seconds is plenty. Those are the "have a think" moments for the kids. 3. If you stumble, don't stop the recording: pause, say the SAME number again, and read that passage again. I'll keep your retake. 4. Kitchen-table warmth beats newsreader polish every time.

Total is roughly two hours of reading — absolutely fine to do over several sittings (each take just needs to start at a numbered item; tick them off on the printed version as you go). When done, drop the audio file(s) — m4a, mp3, wav, anything — into OneDrive Times tables/recordings and tell me.

Tick each number as you record it so you can pick up where you left off.

Part A — Times tables tricks

■ Say: "Number 1" — Double, double

Quick one to start us off — what's 7 times 4? ... If that popped out in under ten seconds, brilliant, you already know this trick. If not, no panic — you're about to learn it right now. Here's the secret: to multiply by 4, you don't multiply at all. You double the number, then double it again. Two easy doubles instead of one hard multiply. Watch it work on 7 times 4. Double 7 is 14. Now double 14 — that's 28. So 7 times 4 is 28. Done, no fuss. Your turn — pause me if you need to: what's 6 times 4? ... And what's 9 times 4? ... Have a proper go before I tell you... Right, let's check together. 6 times 4: double 6 is 12, double 12 is 24. So 6 times 4 is 24. And 9 times 4: double 9 is 18, double 18 is 36. So 9 times 4 is 36. See? Two little doubles and the 4 times table just opens up. Go smash the practice — you've got this one.

■ Say: "Number 2" — Times ten, then halve

Here's a quick challenge — what's 8 times 5? ... Got it in under ten seconds? Then you're already halfway to being a fives master. Took a bit longer? Perfect, because this trick makes it easy every single time. To multiply by 5, times the number by 10 first, then cut your answer in half. That's it — times 10, then halve. Let's walk through 8 times 5. First, 8 times 10 is 80. Now halve it — half of 80 is 40. So 8 times 5 is 40. Easy as that. Your turn — pause me if you need a moment: what's 6 times 5? ... And what's 12 times 5? ... Have a real go before I give the game away... Ready? Let's check them together. 6 times 5: 6 times 10 is 60, and half of 60 is 30. So 6 times 5 is 30. Now 12 times 5: 12 times 10 is 120, and half of 120 is 60. So 12 times 5 is 60. Times 10, then halve — every single time. That's a trick you'll use forever. Nice work.

■ **Say: "Number 3" — Times five, plus one more lot**

Quick fire — what's 7 times 6? ... If you rattled that off in under ten seconds, you're already flying. If it made you pause, stick with me, because this one's about to get much easier. The trick: to multiply by 6, do times 5 first — that's the easy bit — then add one more lot of your number. Times 5, plus one more lot. Let's take 7 times 6 through it. First, 7 times 5 is 35. Now add one more 7: 35 plus 7 is 42. So 7 times 6 is 42. Lovely. Your turn — pause me if you need to: what's 8 times 6? ... And what's 9 times 6? ... Have a proper think before I spill the answers... Let's check them together. 8 times 6: 8 times 5 is 40, add one more 8, and 40 plus 8 is 48. So 8 times 6 is 48. Now 9 times 6: 9 times 5 is 45, add one more 9, and 45 plus 9 is 54. So 9 times 6 is 54. That's it — the sixes just became fives with a bonus. Go show them who's boss.

■ **Say: "Number 4" — Times five, plus times two**

Try this one — what's 8 times 7? ... If that landed in under ten seconds, take a bow, because that's one of the trickiest facts in the whole table. If not, don't worry — most people find the sevens hard, and this trick fixes that. Here's the idea: 7 is just 5 plus 2. So times 7 is times 5 plus times 2 — two easy sums instead of one tough one. Let's do 8 times 7 properly. 8 times 5 is 40. 8 times 2 is 16. Now add them: 40 plus 16 is 56. So 8 times 7 is 56. Your turn — pause me whenever you like: what's 6 times 7? ... And what's 9 times 7? ... Give them a real go first... Right, checking time. 6 times 7: 6 times 5 is 30, and 6 times 2 is 12. Then 30 plus 12 is 42. So 6 times 7 is 42. Now 9 times 7: 9 times 5 is 45, and 9 times 2 is 18. Then 45 plus 18 is 63. So 9 times 7 is 63. See? You've just tamed the sevens — that's a trick plenty of grown-ups never learn.

■ Say: "Number 5" — Double three times

Speedy one — what's 6 times 8? ... Under ten seconds? You're a natural. Slower than that? No bother at all — this trick turns the eights into something you can do half asleep. To multiply by 8, you double the number three times in a row. That's the whole trick. Double, double, double. Let's run 6 times 8 through it. Double 6 is 12. Double 12 is 24. Double 24 is 48. So 6 times 8 is 48 — three quick doubles and you're home. Your turn — pause me if you need thinking time: what's 7 times 8? ... And what's 9 times 8? ... Have a proper crack at both before I tell you... Let's check them together. 7 times 8: double 7 is 14, double 14 is 28, double 28 is 56. So 7 times 8 is 56. Now 9 times 8: double 9 is 18, double 18 is 36, double 36 is 72. So 9 times 8 is 72. Double, double, double — that's all the eights ever ask of you. Go and have fun with the practice, you've earned it.

■ Say: "Number 6" — Nine fingers

Quick test — what's 7 times 9? ... If you nailed it in under ten seconds, superb. If not, get ready, because the answer to every single nine is hiding in your hands. Hold up all 10 fingers. To multiply a number by 9, bend down that number finger — counting from the left. Then just read your hands: fingers to the left of the bent one are the tens, fingers to the right are the ones. Let's do 7 times 9. Bend down finger number 7. Look left — 6 fingers standing. Look right — 3 fingers standing. 6 and 3 — that's 63. So 7 times 9 is 63. Your turn — hands up, pause me if you need to: what's 4 times 9? ... And what's 8 times 9? ... Try it on your fingers first... Let's check. 4 times 9: bend down finger number 4. That leaves 3 on the left and 6 on the right — 3 and 6, so 36. And 8 times 9: bend finger number 8. That's 7 on the left, 2 on the right — 72. So 8 times 9 is 72. Your hands know the whole nine times table — how brilliant is that?

■ **Say: "Number 7" — Times ten, minus one lot**

Here's your challenge — what's 6 times 9? ... Cracked it in under ten seconds? You're already ahead of the game. If it took longer, that's absolutely fine — this trick makes the nines a doddle. The idea: 9 is just one less than 10. So to multiply by 9, do times 10 first, then take one lot of your number away. Let's push 6 times 9 through it. 6 times 10 is 60. Now take away one 6: 60 minus 6 is 54. So 6 times 9 is 54. Two steps, both easy. Your turn — pause me if you want a moment: what's 7 times 9? ... And what's 8 times 9? ... Have a genuine go before the answers arrive... Right, let's check them together. 7 times 9: 7 times 10 is 70, take away one 7, and 70 minus 7 is 63. So 7 times 9 is 63. Now 8 times 9: 8 times 10 is 80, take away one 8, and 80 minus 8 is 72. So 8 times 9 is 72. Times 10, minus one lot — and suddenly the nines are on your side. Lovely work.

■ **Say: "Number 8" — Times ten**

Lightning round — what's 7 times 10? ... If you said 70 before I'd even finished asking, congratulations, you might already own the easiest trick in the whole book. If you hesitated, you'll never hesitate again after this. Times 10 works like this: just put a 0 on the end of the number. What's really happening is every digit slides up one place, and a 0 fills the gap — but all you have to do is pop that 0 on. So for 7 times 10: take the 7, put a 0 on the end, and you get 70. That's genuinely the whole thing. Your turn — pause me if you like: what's 9 times 10? ... And here's a bigger one — what's 34 times 10? ... Say them out loud before I do... Let's check together. 9 times 10: put a 0 on the end of 9 and you get 90. And 34 times 10: put a 0 on the end of 34 — the 3 and the 4 slide up a place — and you get 340. So 34 times 10 is 340. Easiest trick in the book, and now it's yours. Off you go.

■ Say: "Number 9" — Times eleven, single digit

Quick one — what's 7 times 11? ... If that flew out in under ten seconds, you've clearly met this trick before. If not, you're about to love it, because it might be the friendliest trick going. To multiply a single digit by 11, you just write the digit twice. That's it — the number looks in the mirror and there's your answer. So for 7 times 11: take the 7, write it twice — 7 and 7 — and you get 77. So 7 times 11 is 77. It really is that quick. Your turn — pause me if you need to: what's 4 times 11? ... And what's 9 times 11? ... Say them before I do — I bet you can... Let's check together. 4 times 11: write the 4 twice — 4 and 4 — that's 44. So 4 times 11 is 44. And 9 times 11: write the 9 twice — 9 and 9 — that's 99. So 9 times 11 is 99. The number simply looks in the mirror, and the elevens are done. See? You've got this one — go and dazzle someone with it.

■ Say: "Number 10" — Times eleven, two digits

Quick challenge — what's 35 times 11? ... Tricky-sounding, isn't it? If you got it in under ten seconds, you're already a times-11 wizard. If not, stick with me, because this one feels like actual magic. For a two-digit number times 11, add the two digits together and slot the answer in between them. That's it — the outside digits stay put, and their sum squeezes into the middle. Let's do 35 times 11. The digits are 3 and 5. 3 plus 5 is 8. Slot the 8 between the 3 and the 5 — 3, 8, 5. So 35 times 11 is 385. Your turn — pause me if you need to: what's 26 times 11? ... and what's 57 times 11? ... Careful with the second one — have a proper go before I tell you. Ready? 26 times 11: 2 plus 6 is 8, slot it in — 2, 8, 6 — that's 286. Now 57 times 11: 5 plus 7 is 12, and that's ten or more, so the 2 goes in the middle and the 1 carries onto the 5, making 6 — 6, 2, 7 — that's 627. See? You just did a trick most grown-ups have never even heard of.

■ **Say: "Number 11" — Times ten plus times two**

Quick one — what's 6 times 12? ... If that popped out in a few seconds, brilliant, you already own this trick. If it didn't, no worries at all — you're about to learn the easy way. Here's the secret: 12 is just 10 plus 2. So times 12 means times 10 plus times 2 — two sums you already find easy, stuck together. Let's walk through 6 times 12. First, 6 times 10 is 60. Next, 6 times 2 is 12. Now add them: 60 plus 12 is 72. So 6 times 12 is 72. No twelve times table needed — just tens and twos. Your turn — pause me if you like: what's 7 times 12? ... and what's 9 times 12? ... Have a real go before I give the game away... Right, let's check together. 7 times 12: 7 times 10 is 70, 7 times 2 is 14, and 70 plus 14 is 84. And 9 times 12: 9 times 10 is 90, 9 times 2 is 18, and 90 plus 18 is 108. That's it — you've just cracked the twelves. Go smash the practice.

■ **Say: "Number 12" — Times twenty-five**

Here's a spicy one — what's 8 times 25? ... If you rattled that off in under ten seconds, you already think like a mathematician. If not, hang about, because this trick makes it easy every time. The secret is that 25 is a quarter of 100. So to times by 25, times by 100 first — dead easy — then divide by 4. Watch it work on 8 times 25. 8 times 100 is 800. Now divide by 4: 800 divided by 4 is 200. So 8 times 25 is 200. Think of it like money — 8 coins worth 25 pence each make 2 pounds. Your turn — pause me if you need thinking time: what's 12 times 25? ... and what's 6 times 25? ... Have a proper go before I spill the answers... Okay, checking together. 12 times 25: 12 times 100 is 1200, and 1200 divided by 4 is 300. So 12 times 25 is 300. Now 6 times 25: 6 times 100 is 600, and 600 divided by 4 is 150. So 6 times 25 is 150. See? Twenty-fives are basically pocket money maths — you've got this one.

■ Say: "Number 13" — Times fifty

Try this one on for size — what's 6 times 50? ... Got it in a flash? Then you're already halfway to being a human calculator. If it took a while, perfect — this trick is about to make it easy forever. Here's the idea: 50 is half of 100. So times 50 means times 100 first, then halve the answer. Watch. 6 times 50: first, 6 times 100 is 600. Now halve it: half of 600 is 300. So 6 times 50 is 300. Two baby steps instead of one big scary multiply. Your turn — pause me whenever you like: what's 8 times 50? ... and what's 14 times 50? ... Don't peek — have a genuine go first... Right, let's check them together. 8 times 50: 8 times 100 is 800, and half of 800 is 400. So 8 times 50 is 400. Now 14 times 50: 14 times 100 is 1400, and half of 1400 is 700. So 14 times 50 is 700. And that's the whole trick — times 100, cut it in half. You now do fifties faster than most grown-ups. Off you go and prove it.

■ Say: "Number 14" — Times ninety-nine

Big one coming — what's 7 times 99? ... If you nailed that in under ten seconds, take a bow. If your brain went a bit fuzzy, that's completely normal — and this trick will un-fuzz it in about thirty seconds. Here's the thinking: 99 is just one less than 100. So times 99 means times 100 — easy — then take away one lot of your number. Let's run 7 times 99 through it. 7 times 100 is 700. Now take away one 7: 700 minus 7 is 693. So 7 times 99 is 693. Looks like genius maths — it's really just one takeaway. Your turn — pause me if you need a beat: what's 4 times 99? ... and what's 8 times 99? ... Have a proper crack before I tell you... Let's check together. 4 times 99: 4 times 100 is 400, take away one 4, and 400 minus 4 is 396. So 4 times 99 is 396. Now 8 times 99: 8 times 100 is 800, take away one 8, and 800 minus 8 is 792. So 8 times 99 is 792. See? Ninety-nines aren't scary — they're just hundreds in disguise. Go show them who's boss.

■ Say: "Number 15" — Times a hundred

Speed round — what's 7 times 100? ... If you said 700 before I finished asking, you're flying. If you hesitated, no bother — this is honestly the friendliest trick in the whole book. To times by 100, just put two zeros on the end. That's the whole trick. What's really happening is every digit jumping up two places, and the zeros filling the gaps behind it. So for 7 times 100: take the 7, pop two zeros on the end, and you get 700. Done. It works for big numbers too — 42 times 100 is 4200, four thousand two hundred, just like that. Your turn — pause me if you want a moment: what's 9 times 100? ... and what's 36 times 100? ... Say them out loud before I do... Let's check together. 9 times 100: take the 9, add two zeros — 900. And 36 times 100: take the 36, add two zeros — 3600, that's three thousand six hundred. Every digit slid up two places and the zeros filled in behind. That's it — probably the quickest trick you'll ever learn, and you'll use it your whole life. Nice work.

■ Say: "Number 16" — Round to a tidy number, then adjust

Right, a proper brain-stretcher — what's 6 times 29? ... If you got 174 in under ten seconds, honestly, hats off. If your brain said no thanks, that's exactly why this trick exists. Here's the idea: when a number is close to something tidy, use the tidy number instead, then fix it afterwards. 29 is awkward, but 30 is lovely. Watch. 6 times 29: pretend it's 6 times 30, which is 180. But hang on — we used one extra lot of 6, so take a 6 back: 180 minus 6 is 174. So 6 times 29 is 174. Round up, multiply, pay it back. Your turn — pause me if you need to: what's 4 times 39? ... and what's 7 times 19? ... Have a proper go before I jump in... Let's check together. 4 times 39: 39 is nearly 40, and 4 times 40 is 160. We used one extra 4, so take it back: 160 minus 4 is 156. Now 7 times 19: 19 is nearly 20, and 7 times 20 is 140. Take back the extra 7: 140 minus 7 is 133. See? Awkward numbers can't hide near tidy ones any more — you've got this.

■ Say: "Number 17" — Split into tens

Quick fire — what's 7 times 40? ... If that was out of your mouth in a couple of seconds, superb, you already feel this trick in your bones. If not, don't worry — it's a two-step wonder. When you multiply by a round number like 40, split it: do the small times first, then add the zero back on. So times 40 is really times 4, with a zero stuck on the end. Watch it on 7 times 40. First, 7 times 4 is 28. Now add the zero: 280. So 7 times 40 is 280, two hundred and eighty. Big scary forties become friendly little fours. Your turn — pause me if you need a think: what's 6 times 30? ... and what's 8 times 70? ... Have a real go before I say anything... Let's check together. 6 times 30: 6 times 3 is 18, add the zero, and you get 180. Now 8 times 70: 8 times 7 is 56, add the zero, and you get 560, five hundred and sixty. That's the whole move — shrink it, multiply, add the zero back. You now eat round numbers for breakfast. Go get the practice done.

■ Say: "Number 18" — Split by place value

Here's a chunky one — what's 6 times 23? ... If you cracked that in under ten seconds, you're seriously sharp. If it felt too big, brilliant — because this trick chops big numbers down to size. For a big number, break it into tens and ones, multiply each bit on its own, then add the answers up. Let's do 6 times 23. Split 23 into 20 and 3. First bit: 6 times 20 is 120. Second bit: 6 times 3 is 18. Now add them: 120 plus 18 is 138. So 6 times 23 is 138. One hard sum turned into two easy ones. Your turn — pause me if you need to: what's 4 times 32? ... and what's 7 times 24? ... Give them a proper go before I chime in... Let's check together. 4 times 32: split 32 into 30 and 2. 4 times 30 is 120, and 4 times 2 is 8. Add them: 120 plus 8 is 128. Now 7 times 24: split 24 into 20 and 4. 7 times 20 is 140, and 7 times 4 is 28. Add them: 140 plus 28 is 168. See? Split it, smash each piece, stick it back together — you've got this one for life.

■ Say: "Number 19" — Squaring a number ending in five

Quick challenge — what's 35 times 35? ... If that popped out in under ten seconds, you're already a squaring wizard. If not, stick with me, because this one feels like actual magic. Any number ending in 5 is easy to square. Take the front part, multiply it by one more than itself, then stick 25 on the end. That's it. Let's do 35 squared together. The front part is 3. One more than 3 is 4, and 3 times 4 is 12. Now stick 25 on the end of the 12 ... 1225. So 35 times 35 is 1225. Your turn — pause me if you need to: what's 25 squared? ... and what's 45 squared? ... Have a proper go before I tell you... Right, let's check. 25 squared: the front is 2, and 2 times 3 is 6. Stick 25 on the end ... 625. Now 45 squared: the front is 4, and 4 times 5 is 20. Stick 25 on the end ... 2025. So 45 times 45 is 2025. See? Looks like magic — but it's maths, and now it's yours. Go smash the practice.

■ Say: "Number 20" — Squaring a round ten

Here's a big-sounding one — what's 70 times 70? ... Got it in a flash? Then you already own this trick. If it felt scary, don't worry — it's about to become one of your easiest. To square a round number like 70, just square the front digit, then stick two zeros on the end. Two tiny steps. Watch it work on 70 squared. The front digit is 7. 7 times 7 is 49. Now stick two zeros on ... 4900. So 70 times 70 is 4900. A huge sum, done with one little times-table fact. Your turn — pause me if you need to: what's 40 squared? ... and what's 90 squared? ... Have a real go before I say anything... Ready? Let's check together. 40 squared: the front digit is 4, and 4 times 4 is 16. Two zeros on the end ... 1600. So 40 times 40 is 1600. Now 90 squared: 9 times 9 is 81. Two zeros on ... 8100. So 90 times 90 is 8100. That's it — giant squares from tiny facts. You've got this one completely.

■ Say: "Number 21" — Square via the next round number

Try this monster — what's 29 times 29? ... If you cracked that in ten seconds, take a bow. If your brain went blank, perfect — this trick was made for you. To square a number just under a tidy one, use this: the square equals one less times one more, plus 1. So instead of the awkward number, you multiply its two neighbours and add 1. Let's do 29 squared. The neighbours of 29 are 28 and 30. 30 times 28 is 840 — that's just 3 times 28 with a zero on. Now add 1 ... 841. So 29 times 29 is 841. Your turn — pause me if you need to: what's 19 squared? ... and what's 39 squared? ... Have a proper try first... Let's check them. 19 squared: neighbours are 18 and 20. 20 times 18 is 360. Add 1 ... 361. So 19 times 19 is 361. Now 39 squared: neighbours are 38 and 40. 40 times 38 is 1520. Add 1 ... 1521. So 39 times 39 is 1521. See? You just squared numbers most grown-ups would reach for a calculator on.

■ Say: "Number 22" — Both numbers end in five

Deep breath — what's 45 times 25? ... If you rattled that off in ten seconds, you're seriously sharp. If not, brilliant — you're about to learn a proper showpiece. When both numbers end in 5, do this: multiply the front digits, then add half the sum of the fronts, and finish by sticking 25 on the end. Watch it on 45 times 25. The fronts are 4 and 2. 4 times 2 is 8. Now the fronts added: 4 plus 2 is 6, and half of 6 is 3. 8 plus 3 is 11. Stick 25 on the end ... 1125. So 45 times 25 is 1125. Your turn — pause me if you need to: what's 35 times 15? ... and what's 55 times 35? ... Give them a real go... Checking time. 35 times 15: fronts are 3 and 1, and 3 times 1 is 3. Fronts add to 4, half is 2. 3 plus 2 is 5. Stick 25 on ... 525. Now 55 times 35: 5 times 3 is 15. Fronts add to 8, half is 4. 15 plus 4 is 19. Stick 25 on ... 1925. That's it — you now know a trick most grown-ups don't.

■ Say: "Number 23" — Teens times teens

Quick one — what's 13 times 14? ... Under ten seconds? You're flying. Took longer? No worries at all — teens times teens has a lovely shortcut. Here's how it goes: add the ones digit of one number onto the other number, times that by 10, then add the product of the two ones digits. Let's walk 13 times 14 through it. The ones digits are 3 and 4. Take 13 and add the 4 ... 17. 17 times 10 is 170. Now multiply the ones: 3 times 4 is 12. 170 plus 12 is 182. So 13 times 14 is 182. Your turn — pause me if you need to: what's 12 times 15? ... and what's 14 times 16? ... Have a proper go before I tell you... Let's check together. 12 times 15: 12 plus 5 is 17, and 17 times 10 is 170. The ones: 2 times 5 is 10. 170 plus 10 is 180. So 12 times 15 is 180. Now 14 times 16: 14 plus 6 is 20, and 20 times 10 is 200. The ones: 4 times 6 is 24. 200 plus 24 is 224. See? Teens don't scare you any more — go smash the practice.

■ Say: "Number 24" — Same tens digit

Here's a tough-looking one — what's 23 times 26? ... If you nailed it inside ten seconds, hats off. If your eyes went wide, stay with me — there's a neat route through. When both numbers share the same tens digit, use that round tens number as your base. Add the ones digit of one number onto the other, multiply by the base, then add the ones digits multiplied together. Let's do 23 times 26 — the base is 20. Take 23 and add the 6 ... 29. 29 times 20 is 580. Now the ones: 3 times 6 is 18. 580 plus 18 is 598. So 23 times 26 is 598. Your turn — pause me if you need to: what's 21 times 24? ... and what's 32 times 34? ... Have a real crack first... Let's check. 21 times 24: base 20. 21 plus 4 is 25, and 25 times 20 is 500. Ones: 1 times 4 is 4. 500 plus 4 is 504. Now 32 times 34: base 30. 32 plus 4 is 36, and 36 times 30 is 1080. Ones: 2 times 4 is 8. 1080 plus 8 is 1088. Brilliant — that's two-digit multiplying without a pencil in sight.

■ **Say: "Number 25" — Same tens, ones add to ten**

Ready for a whopper? What's 43 times 47? ... If you said 2021 within ten seconds, you might already know the secret. If not — oh, you're going to love this. Here's a beauty: when the tens digits match and the ones digits add to 10, multiply the tens digit by one more than itself for the front of your answer, then multiply the ones digits for the back. Write the two parts together and you're done. Watch 43 times 47. The tens digit is 4, and 4 times 5 is 20 — that's the front. The ones are 3 and 7, and 3 times 7 is 21 — that's the back. Put them together: 20, then 21 ... 2021. So 43 times 47 is 2021. Instant! Your turn — pause me if you need to: what's 62 times 68? ... and what's 24 times 26? ... Give them a proper go... Let's check. 62 times 68: tens digit 6, and 6 times 7 is 42. Ones: 2 times 8 is 16. Together ... 4216. Now 24 times 26: 2 times 3 is 6, and 4 times 6 is 24. Together ... 624. See? You've got a party trick for life now.

■ **Say: "Number 26" — Tens add to ten, same ones**

Try this beast — what's 46 times 66? ... If that took you under ten seconds, I'm impressed. If not, good news: there's a twin trick that makes it easy. This one works when the tens digits add to 10 and the ones digits match. Multiply the tens digits together and add one lot of the ones digit — that's the front of your answer. Then square the ones digit — that's the back. Watch 46 times 66. The tens are 4 and 6, and 4 times 6 is 24. Add the ones digit, 6 ... 30. That's the front. Now square the ones: 6 times 6 is 36. Write them together: 30, then 36 ... 3036. So 46 times 66 is 3036. Your turn — pause me if you need to: what's 34 times 74? ... and what's 47 times 67? ... Have a proper go before I spill the answers... Checking. 34 times 74: tens 3 and 7, and 3 times 7 is 21. Add the ones digit, 4 ... 25. Ones squared: 4 times 4 is 16. Together ... 2516. Now 47 times 67: 4 times 6 is 24, plus 7 is 31. Then 7 times 7 is 49. Together ... 3149. That's it — you now know a trick most grown-ups don't.

■ Say: "Number 27" — Close to one hundred

Last-boss question — what's 97 times 94? ... If you got that in under ten seconds, you're basically a calculator. If not, don't panic — this ancient trick makes it feel easy. It's called Nikhilam, and it works for numbers just under 100. Find how far each number is from 100 — those are the gaps. Cross-subtract: take one number and subtract the other number's gap — that's the front of your answer. Then multiply the two gaps for the back. Let's do 97 times 94. The gap for 97 is 3, and the gap for 94 is 6. Cross-subtract: 97 minus 6 is 91 — the front. Gaps multiplied: 3 times 6 is 18 — the back. Together ... 9118. So 97 times 94 is 9118. Your turn — pause me if you need to: what's 96 times 93? ... and what's 98 times 95? ... Have a real go first... Let's check. 96 times 93: gaps are 4 and 7. 96 minus 7 is 89. Gaps: 4 times 7 is 28. Together ... 8928. Now 98 times 95: gaps are 2 and 5. 98 minus 5 is 93. Gaps: 2 times 5 is 10. Together ... 9310. An ancient Indian trick — and now it lives in your head.

■ Say: "Number 28" — Close to ten

Quick fire — what's 8 times 7? ... If that popped out in a flash, brilliant, you already own this one. If it didn't — no worries at all, here comes a lovely trick. It's the cross trick, and it works for numbers near 10. See how far each number is from 10 — those are the gaps. Cross-subtract one gap from the other number to get your tens, then multiply the gaps together to get your ones. Watch it work on 8 times 7. 8 is 2 away from 10, and 7 is 3 away. Cross-subtract: 8 minus 3 is 5 — that's your tens, so 50. Multiply the gaps: 2 times 3 is 6 — that's your ones. Put them together: 56. Your turn — pause me if you need to: what's 9 times 7? ... and what's 8 times 6? ... Have a proper go before I tell you... Right, let's check. 9 times 7: the gaps are 1 and 3. 9 minus 3 is 6, and 1 times 3 is 3 — so 63. Now 8 times 6: the gaps are 2 and 4. 8 minus 4 is 4, and 2 times 4 is 8 — so 48. See? Two crosses and you're done — go smash the practice.

■ Say: "Number 29" — Square the middle, minus the gap squared

Here's a sneaky one — what's 18 times 22? ... If you cracked that in ten seconds, take a bow. If not, stick with me, because this trick makes it almost instant. When two numbers sit evenly around a nice middle number, square the middle, then subtract the square of the gap. That's the whole trick. Look at 18 times 22. They sit evenly around 20 — one is 2 below, one is 2 above. 20 squared is 400. The gap is 2, and 2 squared is 4. 400 minus 4 is 396. So 18 times 22 is 396. Magic. Your turn — pause me if you need to: what's 29 times 31? ... and what's 17 times 23? ... Have a proper think before I spill the answers... Ready? 29 times 31 sits around 30. 30 squared is 900. The gap is 1, and 1 squared is 1. 900 minus 1 is 899. And 17 times 23 sits around 20. 20 squared is 400. The gap is 3, and 3 squared is 9. 400 minus 9 is 391. That's it — you now know a trick most grown-ups have never even heard of.

■ Say: "Number 30" — One step from a square

Quick one — what's 7 times 8? ... Got it in a blink? Then you already know today's secret. Took a while? Perfect — this trick is about to make it easy forever. Here it is: if you know your square numbers, the neighbours come free. A number times its next-door neighbour is just the square, plus or minus one step. Watch. 7 times 8 — start from 7 squared, which is 49, then add one more 7: 49 plus 7 is 56. Or come from the other side: 8 squared is 64, take away one 8, and 64 minus 8 is 56 again. Same answer both ways. Your turn — pause me if you need to: what's 6 times 7? ... and what's 8 times 9? ... Have a proper go before I tell you... Let's check together. 6 times 7: 6 squared is 36, add one more 6 — 36 plus 6 is 42. And 8 times 9: 8 squared is 64, add one more 8 — 64 plus 8 is 72. Or 9 squared is 81, minus 9 — 72 again. Every square you know unlocks its neighbours. See? You've got this one — go show it off.

■ Say: "Number 31" — Halve and double

Try this monster — what's 16 times 25? ... If you rattled that off in under ten seconds, you're already flying. If your brain went blank — no worries at all, this trick turns monsters into kittens. It's called halve and double. If one number is even, halve it, and double the other one — the answer stays exactly the same. Do it again and again until the sum looks friendly. Watch 16 times 25. Halve 16 to get 8, double 25 to get 50. Now it's 8 times 50, which is 400. Fancy another round? Halve 8 to get 4, double 50 to get 100 — 4 times 100 is 400. Same answer, much kinder numbers. Your turn — pause me if you need to: what's 12 times 15? ... and what's 18 times 35? ... Have a proper go before I jump in... Checking time. 12 times 15: halve 12 to get 6, double 15 to get 30 — 6 times 30 is 180. And 18 times 35: halve 18 to get 9, double 35 to get 70 — 9 times 70 is 630. See? Squash and stretch, and the hard sums melt away — go smash the practice.

■ Say: "Number 32" — Regroup the factors

Speedy one for you — what's 18 times 5? ... Under ten seconds? You legend. Longer? Stick around, because this trick does the heavy lifting for you. The idea: break a number into friendlier factors, then regroup them so a pair makes 10 or 100. Tens and hundreds do all the work. Let's run 18 times 5 through it. 18 is 9 times 2. So the sum is really 9 times 2 times 5 — and look, 2 times 5 is 10. Now it's just 9 times 10, which is 90. So 18 times 5 is 90. Your turn — pause me if you need to: what's 16 times 5? ... and what's 25 times 8? ... Have a proper go before I give the game away... Let's check them together. 16 times 5: split 16 into 8 times 2, then 2 times 5 is 10, and 8 times 10 is 80. Now 25 times 8: split 8 into 4 times 2, then 25 times 4 is 100, and 100 times 2 is 200. Spot the pairs that make 10 or 100 and the sum practically solves itself. That's it — another trick in your back pocket. Go use it.

■ Say: "Number 33" — The two-hand trick for six to ten

Quick challenge — what's 7 times 8? ... If that took you no time at all, lovely. If it's one of those sums that always wobbles — good news, the answer is literally in your hands. Number your fingers from 6 to 10 on each hand, with the thumbs as 6. To multiply, touch the two fingers for your numbers together. Count the touching fingers and everything below — that's your tens. Then multiply the leftover fingers above, one hand times the other — that's your ones. For 7 times 8: touch the 7 finger to the 8 finger. Touching plus below makes 5 fingers, so that's 50. Above, there are 3 fingers on one hand and 2 on the other: 3 times 2 is 6. 50 plus 6 is 56. Your turn — hands ready, pause me if you need to: what's 6 times 9? ... and what's 8 times 8? ... Have a proper go before I check... Here we go. 6 times 9: touching plus below is 1 plus 4 — 5 fingers, so 50. Above: 4 times 1 is 4. 50 plus 4 is 54. And 8 times 8: touching plus below is 3 plus 3 — 6 fingers, so 60. Above: 2 times 2 is 4. 60 plus 4 is 64. Your hands carry this table everywhere — brilliant, isn't it?

Part B — Addition tricks

■ Say: "Number 34" — Left to Right

Quick one to warm up — what's 34 plus 52? ... If you had it before I finished asking, superb. If not, don't panic — this trick sorts it in two easy moves. It's called Left to Right, and it means adding the biggest parts first. Split each number into tens and ones, add the tens, add the ones, then put the two answers together. Watch 34 plus 52. The tens first: 30 plus 50 is 80. Now the ones: 4 plus 2 is 6. Put them together: 80 plus 6 is 86. So 34 plus 52 is 86. Always start with the tens — that's the big chunk of the answer done straight away. Your turn — pause me if you need to: what's 45 plus 23? ... and what's 61 plus 36? ... Have a proper go before I tell you... Let's check them. 45 plus 23: tens first, 40 plus 20 is 60. Ones next, 5 plus 3 is 8. Together: 60 plus 8 is 68. And 61 plus 36: 60 plus 30 is 90, then 1 plus 6 is 7, and 90 plus 7 is 97. See? Big bits first and the sum falls open — go smash the practice.

■ Say: "Number 35" — Number Bonds

Lightning round — what goes with 7 to make 10? ... If you said 3 before you could blink, you already know today's trick. If you had to count on your fingers — perfect, let's fix that right now. Number bonds are the pairs that make 10: 7 plus 3, 6 plus 4, 8 plus 2, 5 plus 5. The trick is simple — learn them off by heart so you never have to count them out. And here's the bonus: they work for 100 too. 70 plus 30, 60 plus 40, 80 plus 20 — the same pairs, just ten times bigger. So for our opener: 7 plus 3 is 10 — the missing partner was 3, instantly, no counting. Your turn — pause me if you need to: what goes with 6 to make 10? ... and 70 plus what makes 100? ... Say them out loud before I do... Checking time. 6 needs its partner 4, because 6 plus 4 is 10. And 70 needs 30, because 70 plus 30 is 100 — that's just the bond 7 plus 3 wearing a bigger coat. Knowing your bonds makes every other trick in this game faster. That's it — you've got the building blocks now. Go play.

■ Say: "Number 36" — Bigger Number First

Quick one — what's 3 plus 28? ... If you got 31 in a couple of seconds, nice work. If you started counting up from 3 — ooh, stop right there, because this trick will save you loads of effort. Here's the secret: adding is easier if you start with the bigger number. Flipping the sum round changes nothing about the answer — 3 plus 28 is exactly the same as 28 plus 3. So don't slog up from 3. Start at 28 and count on just three little hops: 29, 30, 31. So 3 plus 28 is 31. Done in a breath. Your turn — pause me if you need to: what's 2 plus 47? ... and what's 4 plus 65? ... Have a proper go before I check... Right then. 2 plus 47: flip it to 47 plus 2, count on — 48, 49. So 2 plus 47 is 49. And 4 plus 65: flip it to 65 plus 4, count on — 66, 67, 68, 69. So 4 plus 65 is 69. Big number first, tiny hops after. See? You've got this one — go race through the practice.

■ Say: "Number 37" — Count On

Quick one — what's 46 plus 3? ... If that popped out in a couple of seconds, you're already doing this trick without knowing it. If not, no bother at all — it takes about a minute to learn. Here it is: when you're adding 1, 2 or 3, don't work anything out. Just count up from the big number, one little hop at a time. A few hops and you're done. So for 46 plus 3, start at 46 and hop: 47 ... 48 ... 49. Three hops, and you land on 49. So 46 plus 3 is 49. Your turn — pause me if you need to: what's 65 plus 2? ... and what's 88 plus 3? ... Have a proper go before I tell you ... Right, let's check together. 65 plus 2: start at 65 and hop twice — 66 ... 67. So 65 plus 2 is 67. Now 88 plus 3: start at 88 — 89 ... 90 ... 91. So 88 plus 3 is 91. See how it hops straight over the 90 without a wobble? That's it — you now know a trick you'll use every single day.

■ Say: "Number 38" — Doubles

Quick one — what's 7 plus 7? ... If you said 14 before I finished asking, brilliant — you already know this one. If you had to count it out, stick with me, because this trick is all about never counting again. A double is just a number added to itself, and the secret is simple: learn them off by heart, so they're instant. Doubles are the anchors that loads of other sums lean on. Let's walk the family: 6 plus 6 is 12. 7 plus 7 is 14. So our hook question, 7 plus 7 — no working out, no fingers — it's just 14, straight from memory. Your turn — pause me if you need to: what's 8 plus 8? ... and what's 9 plus 9? ... Say them out loud before I do ... Ready? 8 plus 8 is 16 — that's 8 added to itself, 16, every time. And 9 plus 9 is 18 — again, no counting, just 18. Say them once more: 16 ... 18. Lovely. The more you say them, the faster they get, until they're as automatic as your own name. See? You've got this one — go smash the practice.

■ Say: "Number 39" — Near Doubles

Quick one — what's 6 plus 7? ... Got it in under ten seconds? Then you're halfway to mastering this already. If not, don't worry a bit — this trick makes it easy. A near double is a sum that's almost a double. So spot the double hiding inside, double the smaller number, then add the extra 1. Let's do 6 plus 7 properly. It's nearly 6 plus 6. Double the 6 to get 12, then add the extra 1 to get 13. So 6 plus 7 is 13. And here's a bonus tip: if the two numbers are 2 apart, just double the middle one. Your turn — pause me if you need to: what's 8 plus 9? ... and what's 7 plus 9? ... Have a proper think before I say ... Let's check together. 8 plus 9: that's nearly 8 plus 8. Double 8 is 16, add the extra 1 — 17. So 8 plus 9 is 17. Now 7 plus 9: those are 2 apart, so double the middle number, which is 8. Double 8 is 16. So 7 plus 9 is 16. That's it — you now know a trick most grown-ups don't.

■ Say: "Number 40" — Make Ten

Quick one — what's 8 plus 5? ... If that came out in a flash, you already know this one. If it didn't, brilliant — you're about to learn my favourite adding trick. It's called Make Ten. Instead of adding straight across, you jump to the next ten first, because tens are easy. Ask how much gets you to 10, split the second number to match, and finish the job. Watch with 8 plus 5. How many gets 8 up to 10? That's 2. So split the 5 into 2 and 3. Now 8 plus 2 is 10, and 10 plus 3 is 13. So 8 plus 5 is 13. Your turn — pause me if you need to: what's 7 plus 8? ... and what's 9 plus 8? ... Have a proper go before I tell you ... Let's check them together. 7 plus 8: 7 needs 3 to make 10, so split the 8 into 3 and 5. 7 plus 3 is 10, then 10 plus 5 is 15. And 9 plus 8: 9 only needs 1, so split the 8 into 1 and 7. 9 plus 1 is 10, then 10 plus 7 is 17. See? You've got this one — go smash the practice.

■ Say: "Number 41" — Spot the Tens

Quick one — what's 6 plus 3 plus 4 plus 7? ... If you rattled that off in under ten seconds, you might already be using this trick. If not, no worries — it's a lovely one. When you're adding a whole string of numbers, don't plod through in order. Scan the line and spot the pairs that make 10 first, because tens are the easiest things in the world to add. Let's do our hook question. In 6 plus 3 plus 4 plus 7, spot the pairs: 6 plus 4 is 10, and 3 plus 7 is 10. Now just add the tens: 10 plus 10 is 20. So 6 plus 3 plus 4 plus 7 is 20. Your turn — pause me if you need to: what's 8 plus 5 plus 2 plus 5? ... and what's 5 plus 9 plus 5? ... Hunt for those pairs before I tell you ... Let's check together. 8 plus 5 plus 2 plus 5: the 8 and the 2 make 10, and 5 plus 5 makes 10 — so 10 plus 10 is 20. And 5 plus 9 plus 5: pair the 5 and 5 for 10, then 10 plus 9 is 19. See? Once you spot the tens, long sums fall apart — go smash the practice.

■ Say: "Number 42" — Round and Adjust

Quick one — what's 38 plus 27? ... If you got that in under ten seconds, hats off — you already think like this. If not, no worries at all, because this trick makes messy sums tidy. It's called Round and Adjust. Round one number up to something tidy, do the easy sum, then fix it afterwards by taking back what you borrowed. Watch with 38 plus 27. Round 38 up to 40 — that's 2 extra. Now the easy sum: 40 plus 27 is 67. Take back the 2 you added: 67 minus 2 is 65. So 38 plus 27 is 65. Your turn — pause me if you need to: what's 49 plus 24? ... and what's 68 plus 15? ... Have a proper go before I tell you ... Let's check together. 49 plus 24: round 49 up to 50 — that's 1 extra. 50 plus 24 is 74, then take back the 1: 74 minus 1 is 73. And 68 plus 15: round 68 up to 70 — that's 2 extra. 70 plus 15 is 85, take back the 2: 85 minus 2 is 83. That's it — you now know a trick most grown-ups don't.

■ Say: "Number 43" — Give and Take

Quick one — what's 47 plus 26? ... Under ten seconds? Then this trick is already living in your head. If not, perfect — let's put it there now. It's called Give and Take, and the idea is lovely: you can give a bit from one number to the other, and the total stays exactly the same. Give just enough to make one number a nice round ten. Watch with 47 plus 26. Give 3 from the 26 across to the 47. The 47 becomes 50, and the 26 becomes 23. Now it's easy: 50 plus 23 is 73. So 47 plus 26 is 73. Your turn — pause me if you need to: what's 58 plus 34? ... and what's 29 plus 45? ... Have a proper go before I tell you ... Right, let's check together. 58 plus 34: give 2 from the 34 to the 58. That makes 60 plus 32, which is 92. So 58 plus 34 is 92. And 29 plus 45: give 1 from the 45 to the 29. That makes 30 plus 44, which is 74. So 29 plus 45 is 74. See? You've got this one — go smash the practice.

■ Say: "Number 44" — Add 9 and 99

Quick one — what's 56 plus 9? ... If you got that in under ten seconds, you already know this one. If not — no worries at all, let's learn it right now. Adding 9 is sneaky-easy, because 9 is nearly 10. So don't add 9 at all: add 10, which is dead easy, then take 1 back. Watch with 56 plus 9. Add 10 to 56 and you get 66. Now take 1 back: 66 minus 1 is 65. So 56 plus 9 is 65. And the same idea works for 99 — add 100, then take 1 back. Your turn — pause me if you need to: what's 47 plus 9? ... and what's 34 plus 99? ... Have a proper go before I tell you ... Let's check them together. 47 plus 9: add 10 to get 57, then take 1 back — 56. So 47 plus 9 is 56. Now 34 plus 99: add 100 to get 134, then take 1 back — 133. So 34 plus 99 is 133. A big scary 99 turned into two baby steps. That's it — you now know a trick most grown-ups don't.

Part C — Subtraction tricks

■ Say: "Number 45" — Take Away in Parts

Quick one — what's 56 minus 23? ... If that came to you in under ten seconds, you're already thinking like a maths whizz. If not, don't worry a bit — this trick sorts it out. It's called Take Away in Parts. Instead of subtracting the whole number in one scary go, take away the big part first — the tens — then the small part, the ones. Watch with 56 minus 23. First take away the 20: 56 minus 20 is 36. Now take away the 3: 36 minus 3 is 33. So 56 minus 23 is 33. Two gentle steps instead of one hard one. Your turn — pause me if you need to: what's 78 minus 45? ... and what's 87 minus 34? ... Have a proper go before I tell you ... Let's check together. 78 minus 45: take away the 40 first — 78 minus 40 is 38. Then take away the 5: 38 minus 5 is 33. So 78 minus 45 is 33. And 87 minus 34: take away the 30 — 87 minus 30 is 57. Then the 4: 57 minus 4 is 53. So 87 minus 34 is 53. See? You've got this one — go smash the practice.

■ Say: "Number 46" — Count Back

Quick one — what's 52 minus 3? ... If that popped out in a couple of seconds, brilliant, you've already got this trick. If you had to think hard, stick with me — it takes about a minute to learn. When you're taking away just 1, 2 or 3, don't do any working out at all. Just count backwards, one little step at a time. Three small hops back beats one big sum every time. Watch it with 52 minus 3. Start at 52 and step back three times: 51 ... 50 ... 49. That's it — 52 minus 3 is 49. Your turn — pause me if you need to: what's 74 minus 2? ... and what's 61 minus 3? ... Have a proper go before I tell you... Right, let's check together. 74 minus 2: start at 74 and hop back twice — 73 ... 72. So 74 minus 2 is 72. Now 61 minus 3: start at 61 and take three steps back — 60 ... 59 ... 58. So 61 minus 3 is 58. See? You've got this one — quick little steps back, and off you go to the practice.

■ Say: "Number 47" — Count Up

Try this one on for size — what's 42 minus 38? ... If you got it in under ten seconds, you're already thinking like a maths whizz. If not, no stress — this trick makes it easy. When the two numbers are close together, don't count backwards at all. Count UP from the smaller number to the bigger one, and count how many steps it takes. That number of steps is your answer. Let's do 42 minus 38. Start at 38 and climb: 39, 40, 41, 42. That took 4 steps, so 42 minus 38 is 4. So much friendlier than counting back 38! Your turn — pause me if you need to: what's 71 minus 66? ... and what's 53 minus 49? ... Have a real go before I spill the answers... Let's check them together. 71 minus 66: start at 66 and count up — 67, 68, 69, 70, 71. That's 5 steps, so 71 minus 66 is 5. And 53 minus 49: start at 49 — 50, 51, 52, 53. That's 4 steps, so 53 minus 49 is 4. That's it — when the numbers are close, climb up instead of falling back. Go smash the practice.

■ Say: "Number 48" — Bridge Down

Here's a quick one — what's 43 minus 8? ... Got it in a flash? Then you're already bridging like a pro. Took a bit longer? Perfect — this trick is exactly for you. It's called Bridge Down, and the idea is simple: step down onto the nearest ten first, because tens are lovely to work from. Split the number you're taking away into two friendly bits. Watch with 43 minus 8. How many gets you from 43 down to 40? That's 3. So split the 8 into 3 plus 5. First jump: 43 minus 3 is 40. Second jump: 40 minus 5 is 35. So 43 minus 8 is 35. Two easy hops with a ten in the middle. Your turn — pause me if you need to: what's 62 minus 5? ... and what's 34 minus 6? ... Have a proper go first... Let's check together. 62 minus 5: from 62 down to 60 is 2, so split 5 into 2 plus 3. 62 minus 2 is 60, then 60 minus 3 is 57. And 34 minus 6: from 34 down to 30 is 4, so split 6 into 4 plus 2. 34 minus 4 is 30, then 30 minus 2 is 28. See? Land on the ten and the rest is easy — you've got this one.

■ Say: "Number 49" — Think Addition

Quick fire — what's 14 minus 9? ... Under ten seconds? You're flying. Longer? No worries at all — here's a sneaky way to make it easy. The secret is that every subtraction is really an addition wearing a disguise. Instead of taking away, flip the question round and ask what you'd need to ADD. You already know loads of addition facts, so let them do the work. So for 14 minus 9, ask it the other way: 9 plus what makes 14? Think... 9 plus 5 is 14. So 14 minus 9 is 5. No counting back, no fuss — just an addition fact you already knew, flipped over. Your turn — pause me if you need to: what's 13 minus 7? ... and what's 16 minus 8? ... Have a proper think before I tell you... Let's check them together. 13 minus 7 — flip it: 7 plus what makes 13? 7 plus 6 is 13, so 13 minus 7 is 6. And 16 minus 8 — flip it: 8 plus what makes 16? That's a double you know: 8 plus 8 is 16, so 16 minus 8 is 8. That's it — you now know a trick most grown-ups never spot. Go show the practice who's boss.

■ Say: "Number 50" — Round and Adjust

Here's a challenge — what's 62 minus 29? ... If you cracked that in under ten seconds, hats off, you probably already use this trick. If it felt awkward, that's the 29's fault, and we're about to fix it. The trick is Round and Adjust: round the number you're taking away to something tidy, do the easy sum, then fix it afterwards. Watch with 62 minus 29. That 29 is nearly 30, so round it up to 30. Now the easy bit: 62 minus 30 is 32. But hang on — we took away 1 too many, so give 1 back: 32 plus 1 is 33. So 62 minus 29 is 33. Your turn — pause me if you need to: what's 84 minus 38? ... and what's 71 minus 19? ... Have a proper go before I tell you... Let's check together. 84 minus 38: round 38 up to 40. 84 minus 40 is 44. We took 2 too many, so add 2 back: 44 plus 2 is 46. And 71 minus 19: round 19 up to 20. 71 minus 20 is 51. Give 1 back: 51 plus 1 is 52. See? Messy numbers turned tidy — you've absolutely got this one.

■ Say: "Number 51" — Same Difference

Quick one — what's 74 minus 37? ... If that came out smoothly in a few seconds, you might already know today's trick. If your brain went "urgh", perfect — this will change everything. It's called Same Difference, and here's the magic: the gap between two numbers doesn't change if you slide BOTH numbers up by the same amount. So slide them until the sum turns tidy. Watch with 74 minus 37. Add 3 to both numbers: 74 becomes 77, and 37 becomes 40. Now it's 77 minus 40, which is 37. So 74 minus 37 is 37. Same gap, much friendlier numbers. Your turn — pause me if you need to: what's 82 minus 58? ... and what's 53 minus 27? ... Have a proper go before I give the game away... Let's check together. 82 minus 58: add 2 to both, so it becomes 84 minus 60. That's 24, so 82 minus 58 is 24. And 53 minus 27: add 3 to both, so it becomes 56 minus 30. That's 26, so 53 minus 27 is 26. See? Slide both numbers, keep the gap — you now know a trick most grown-ups don't. Off you go.

■ Say: "Number 52" — Subtract from 100

Challenge time — what's 100 minus 42? ... If you fired that back in under ten seconds, you already think in pairs that make 100. If not, this trick is a beauty. To subtract from 100, don't subtract at all — find the partner that pairs up with your number to make 100. And here's the shortcut: if the number doesn't end in 0, the ones digits add to 10 and the tens digits add to 9. Watch with 100 minus 42. Ones digit is 2, and 2 needs 8 to make 10. Tens digit is 4, and 4 needs 5 to make 9. So the partner is 58 — and 100 minus 42 is 58. Check it: 42 plus 58 is 100. Lovely. Your turn — pause me if you need to: what's 100 minus 76? ... and what's 100 minus 30? ... Have a proper go first... Let's check together. 100 minus 76: ones — 6 needs 4 to make 10. Tens — 7 needs 2 to make 9. So the answer is 24. And 100 minus 30 ends in 0, so just use tens bonds: 3 plus 7 is 10, so the answer is 70. See? Ones to 10, tens to 9 — you've got this one. Go smash the practice.

■ Say: "Number 53" — Subtract 9 and 99

Quick one — what's 45 minus 9? ... Under ten seconds and you're already ahead of the game. Took longer? No bother — this trick makes 9s a doddle forever. Here's the idea: taking away 9 is nearly the same as taking away 10, and 10 is easy. So take away 10, then give 1 back, because you took 1 too many. Watch with 45 minus 9. Take away 10: 45 minus 10 is 35. Now give 1 back: 35 plus 1 is 36. So 45 minus 9 is 36. And it works for 99 too — take away 100, then give 1 back. Your turn — pause me if you need to: what's 72 minus 9? ... and here's a big one: what's 431 minus 99? ... Have a proper go before I tell you... Let's check together. 72 minus 9: take away 10 to get 62, then give 1 back — 63. So 72 minus 9 is 63. And 431 minus 99: take away 100 to get 331, then give 1 back — 332. So 431 minus 99 is 332. See? Even a big scary sum falls over with this one — go show the practice what you can do.

Part D — Division tricks

■ Say: "Number 54" — Times Tables Backwards

Quick one — what's 42 divided by 6? ... If that popped out in a flash, you already know today's secret. If it didn't, no worries — stick with me for one minute. Here's the trick: division is just multiplication flipped backwards. So instead of dividing, you ask, times what? Every division fact is a times-table fact in disguise. Let's do our question. 42 divided by 6 is really asking: 6 times what makes 42? Run through your sixes... 6 times 7 is 42. So 42 divided by 6 is 7. That's it. Your turn — pause me if you need to: what's 63 divided by 9? ... and what's 32 divided by 8? ... Have a proper go before I tell you... Right, let's check together. 63 divided by 9 asks: 9 times what makes 63? Well, 9 times 7 is 63 — so the answer is 7. And 32 divided by 8 asks: 8 times what makes 32? 8 times 4 is 32 — so the answer is 4. See? If your times tables are sharp, division is instant. You've already done the hard work — go smash the practice.

■ Say: "Number 55" — $\div 10, 100, 1000$

Quick fire — what's 350 divided by 10? ... If you said 35 before I finished asking, brilliant, you know this one. If not, you're about to learn the easiest division trick there is. To divide by 10, just take one zero off the end. To divide by 100, take two zeros off. To divide by 1000, take three off. That's really the decimal point sliding one place to the left for each zero. So, our question: 350 divided by 10. Take one zero off the end of 350... and you're left with 35. So 350 divided by 10 is 35. Your turn — pause me if you need to: what's 6200 divided by 100? ... and what's 81000 divided by 1000? ... Have a real go before I say anything... Let's check together. 6200 divided by 100: dividing by 100 means taking two zeros off. Take two zeros off 6200 and you get 62. And 81000 divided by 1000: three zeros in 1000, so take three zeros off. 81000 becomes 81. That's the whole trick — count the zeros, chop the zeros. You've got this one — go and show it off.

■ Say: "Number 56" — The 2, 5, 10 Test

Here's a challenge — without doing any dividing at all, can you tell me if 465 divides exactly by 5? ... If you spotted it straight away, you're already a detective. If not, here's the secret: you only ever need to peek at the last digit. A number divides by 2 if it ends in 0, 2, 4, 6 or 8 — that's the even numbers. It divides by 5 if it ends in 0 or 5. And it divides by 10 if it ends in 0. So, 465 — look at the last digit. It's a 5. Ends in 5 means yes, it divides by 5. No sums needed at all. Your turn — pause me if you need to: does 738 divide by 2? ... and does 890 divide by 10? ... Have a proper think before I tell you... Let's check together. 738 ends in an 8 — that's even — so yes, it divides by 2. And 890 ends in a 0 — so yes, it divides by 10. And here's a bonus: because 890 ends in 0, it divides by 2 and 5 as well. All three, from one glance. That's it — you now know a trick most grown-ups don't.

■ **Say: "Number 57" — Digit Sum for 3 and 9**

Try this — does 234 divide exactly by 3? No calculator, no long division. ... If you knew a sneaky way to check, take a bow. If not, here it comes. Add up the digits. If the digits add to a multiple of 3, the whole number divides by 3. And if the digits add to a multiple of 9, it divides by 9 too. Let's test 234. Add the digits: 2 plus 3 plus 4 is 9. Is 9 a multiple of 3? Yes — 3 times 3 is 9. So 234 divides by 3. Easy as that. Your turn — pause me if you need to: does 512 divide by 3? ... and does 738 divide by 9? ... Have a proper go before I spill the answers... Let's check together. For 512: 5 plus 1 plus 2 is 8. Is 8 a multiple of 3? No — so 512 does not divide by 3. And for 738: 7 plus 3 plus 8 is 18, and 18 is 9 times 2 — a multiple of 9 — so yes, 738 divides by 9. Adding a few digits beats dividing a big number every time. Go smash the practice — you've earned it.

■ **Say: "Number 58" — $\div 4$ = Halve Twice**

Quick one — what's 84 divided by 4? ... If you got 21 in under ten seconds, you might already know this trick. If not, don't worry — it's a lovely one. Dividing by 4 in one go is hard. But 4 is just 2 times 2 — so dividing by 4 is the same as halving twice. Two easy halves instead of one tricky division. Watch it work on 84 divided by 4. Half of 84 is 42. Half of 42 is 21. So 84 divided by 4 is 21. Two little steps, done in your head. Your turn — pause me if you need to: what's 56 divided by 4? ... and what's 132 divided by 4? ... Have a proper go before I tell you... Right, let's check together. 56 divided by 4: half of 56 is 28, and half of 28 is 14. So 56 divided by 4 is 14. And 132 divided by 4: half of 132 is 66, and half of 66 is 33. So 132 divided by 4 is 33. Halve it, halve it again — that's the whole trick. See? You've got this one — go smash the practice.

■ **Say: "Number 59" — $\div 8 = \text{Halve Three Times}$**

Here's a tough-sounding one — what's 96 divided by 8? ... If you rattled that off, hats off to you. If it made you wince, perfect — this trick is about to make it easy. 8 is 2 times 2 times 2. So dividing by 8 is just halving three times in a row. No big division — three friendly little halves. Let's take 96 divided by 8. Half of 96 is 48. Half of 48 is 24. Half of 24 is 12. So 96 divided by 8 is 12. Halve, halve, halve — done. Your turn — pause me if you need to: what's 88 divided by 8? ... and what's 240 divided by 8? ... Have a real go before I give it away... Let's check together. 88 divided by 8: half of 88 is 44, half of 44 is 22, and half of 22 is 11. So 88 divided by 8 is 11. And 240 divided by 8: half of 240 is 120, half of 120 is 60, and half of 60 is 30. So 240 divided by 8 is 30. Three quick halves beats one scary division every time. That's it — you now know a trick most grown-ups don't.

■ **Say: "Number 60" — $\div 5 = \text{Double then } \div 10$**

Quick challenge — what's 340 divided by 5? ... If you got 68 in a blink, you may already know this gem. If not, you're going to love it. Dividing by 5 feels awkward — but here's the secret: it's exactly the same as doubling the number, then dividing by 10. Why? Because one fifth is the same as two tenths. Doubling is easy, and dividing by 10 just knocks the last zero off — or slides the decimal point one place left. Let's do 340 divided by 5. Double 340: that's 680. Now divide by 10: 68. So 340 divided by 5 is 68. Your turn — pause me if you need to: what's 270 divided by 5? ... and what's 415 divided by 5? ... Have a proper go before I tell you... Let's check together. 270 divided by 5: double 270 is 540, then divide by 10 gives 54. So 270 divided by 5 is 54. And 415 divided by 5: double 415 is 830, then divide by 10 gives 83. So 415 divided by 5 is 83. Double, then chop by 10 — and you can do the two steps in either order. See? You've got this one — go smash the practice.

■ Say: "Number 61" — Last Two for 4, Last Three for 8

Big number alert — does 5324 divide exactly by 4? ... If you answered that without breaking a sweat, you already know the shortcut. If it looked terrifying, good news: you can ignore almost all of it. For dividing by 4, only the last two digits matter — if they divide by 4, the whole number does. For dividing by 8, only the last three digits matter. The front of the number can be as long as you like; you never touch it. So, 5324: the last two digits are 24. Does 24 divide by 4? Yes — 24 divided by 4 is 6. So the whole of 5324 divides by 4. Your turn — pause me if you need to: does 7136 divide by 4? ... and does 9240 divide by 8? ... Have a proper go before I tell you... Let's check together. 7136: last two digits are 36, and 36 divided by 4 is 9 — so yes, 7136 divides by 4. And 9240: last three digits are 240, and 240 divided by 8 is 30 — so yes, 9240 divides by 8. Huge numbers, tiny checks. That's it — you now know a trick most grown-ups don't.

■ Say: "Number 62" — Halve Both First

Quick one — what's 96 divided by 12? ... If you got 8 straight off, lovely — this trick will still speed you up. If you didn't, no worries at all, here it comes. When the number you're dividing and the number you're dividing by are both even, you can halve them both — and the answer stays exactly the same. Keep halving as long as both stay even, and a scary division turns friendly. Let's do 96 divided by 12. Both even, so halve both: 48 divided by 6. Still both even — halve again: 24 divided by 3. That's easy: 24 divided by 3 is 8. So 96 divided by 12 is 8. One warning: if there's a remainder, it got halved every time you halved, so double it back once for each halving at the end. Your turn — pause me if you need to: what's 140 divided by 20? ... and what's 180 divided by 12? ... Have a proper go before I tell you... Let's check together. 140 divided by 20: halve both to get 70 divided by 10, which is 7. And 180 divided by 12: halve both to get 90 divided by 6, halve again to get 45 divided by 3, which is 15. See? You've got this one — go smash the practice.

■ Say: "Number 63" — Split the Dividend

Big challenge this time — what's 864 divided by 6? ... If you got that in under ten seconds, you're already a division wizard. If your brain went "no chance", perfect — this trick is exactly for scary-looking sums like that. The secret: split the big number into friendly chunks that each divide easily, do the little divisions, then add the answers. Watch. 864 divided by 6: split 864 into 600 plus 264. Now, 600 divided by 6 is 100 — easy. And 264 divided by 6 is 44. Add them up: 100 plus 44 is 144. So 864 divided by 6 is 144. Two easy divisions instead of one monster. Your turn — pause me if you need to: what's 848 divided by 4? Try splitting it into 800 plus 48... And what's 396 divided by 3? Try 300 plus 96... Have a proper go... Right, together now. 848 divided by 4: 800 divided by 4 is 200, and 48 divided by 4 is 12. 200 plus 12 is 212. And 396 divided by 3: 300 divided by 3 is 100, and 96 divided by 3 is 32. 100 plus 32 is 132. See? Monster sums, chopped down to size — that's proper mathematician thinking.

■ Say: "Number 64" — Front-End First

Quick one — what's 256 divided by 8? ... If that popped out in under ten seconds, you're already ahead of the game. If not, no stress at all — this trick will sort it. Mental division is easiest from left to right. First ask: about how many tens of the divisor fit? Then deal with whatever's left over. Let's do 256 divided by 8. How many tens? Well, 8 times 30 is 240, and that fits. So the answer starts with 30-something. Now what's left: 256 minus 240 is 16. And 16 divided by 8 is 2. Put them together: 30 plus 2 is 32. So 256 divided by 8 is 32. Your turn — pause me if you need to: what's 192 divided by 6? ... and what's 144 divided by 4? ... Have a proper go before I tell you... Right, let's check. 192 divided by 6: 6 times 30 is 180 — that fits. 192 minus 180 is 12, and 12 divided by 6 is 2. So 30 plus 2 is 32. Now 144 divided by 4: 4 times 30 is 120. 144 minus 120 is 24, and 24 divided by 4 is 6. So 30 plus 6 is 36. See? Big divisions, tamed. Go smash the practice.

■ Say: "Number 65" — Subtract Friendly Chunks

Here's a chunky one — what's 432 divided by 6? ... If you cracked that in under ten seconds, hats off, you barely need me. If not, brilliant — you're about to learn a lovely trick. You build the answer up by subtracting easy multiples of the divisor, starting big. Each time, keep note of how many lots you took. At the end, add up all your friendly multipliers. Let's do 432 divided by 6. Start big: 6 times 50 is 300. Take it off: 432 minus 300 is 132. Next: 6 times 20 is 120. Take it off: 132 minus 120 is 12. Then 6 times 2 is 12. Take it off: 12 minus 12 is 0. Now add the multipliers: 50 plus 20 plus 2 is 72. So 432 divided by 6 is 72. Your turn — pause me if you need to: what's 345 divided by 5? ... and what's 288 divided by 4? ... Give it a real go first... Checking time. 345 divided by 5: 5 times 50 is 250, and 345 minus 250 is 95. 5 times 10 is 50, and 95 minus 50 is 45. 5 times 9 is 45, leaving 0. Add them: 50 plus 10 plus 9 is 69. And 288 divided by 4: 4 times 50 is 200, leaving 88. 4 times 20 is 80, leaving 8. 4 times 2 is 8, leaving 0. So 50 plus 20 plus 2 is 72. That's it — chunk by chunk, you can divide anything.

■ Say: "Number 66" — Round to Estimate

Quick one — roughly what's 347 divided by 6? Not exactly — just roughly. ... If you said about 60 within ten seconds, you already think like a mathematician. If not, don't worry — this is one of the most useful habits in all of maths. When the numbers are awkward, round them to something tidier before you start. Pick a nearby number that divides cleanly, and you'll instantly know the size of the answer — which stops silly mistakes. Let's do 347 divided by 6. 347 is awkward, but 360 is a multiple of 6, and it's nearby. 360 divided by 6 is 60. So 347 divided by 6 is about 60 — a bit less, actually, since 347 is a bit under 360. Your turn — pause me if you need to: roughly what's 415 divided by 7? ... and roughly what's 251 divided by 5? ... Have a proper think before I tell you... Let's check together. 415 divided by 7: round 415 up to 420, because 420 divided by 7 is 60. So the answer is about 60. And 251 divided by 5: round 251 down to 250, and 250 divided by 5 is 50. So it's just over 50. Always estimate before a big division — you'll spot a wrong answer a mile off. You've got this one.

■ Say: "Number 67" — The 11 Test

Tricky one — does 2915 divide exactly by 11? ... If you knew that in under ten seconds, you might be part calculator. If not, perfect — this trick feels like actual magic. To test for 11, you add and subtract the digits alternately: plus, minus, plus, minus. If the answer is 0 or any multiple of 11, the number divides by 11. Let's test 2915. The digits are 2, 9, 1, 5. Go alternately: 2 minus 9 is minus 7 ... plus 1 is minus 6 ... minus 5 is minus 11. Minus 11 is a multiple of 11 — so yes! 2915 divides by 11 exactly: it's 11 times 265. Your turn — pause me if you need to: does 253 divide by 11? ... and does 175 divide by 11? ... Have a proper go with the plus-minus pattern... Let's check. For 253: 2 minus 5 is minus 3, plus 3 is 0. Zero counts — so yes, 253 divides by 11. It's 11 times 23. For 175: 1 minus 7 is minus 6, plus 5 is minus 1. Minus 1 is not a multiple of 11 — so no, 175 does not divide by 11. See? You've just learned a test most grown-ups have never even heard of.

■ Say: "Number 68" — The 7 Trick

Here's a sneaky one — does 364 divide exactly by 7? ... If you spotted that in under ten seconds, your 7 times table is in great shape. If not, no worries — there's a secret test for sevens. Take the last digit, double it, and subtract that from the rest of the number. If what's left divides by 7, so does the whole number. Let's test 364. The last digit is 4, and double 4 is 8. The rest of the number is 36. Now 36 minus 8 is 28 — and 28 divided by 7 is 4, so yes! 364 divides by 7. In fact, 364 divided by 7 is 52. Your turn — pause me if you need to: does 315 divide by 7? ... and does 249 divide by 7? ... Try the double-and-subtract test before I tell you... Let's check. For 315: last digit 5, double it to get 10. The rest is 31, and 31 minus 10 is 21. 21 divided by 7 is 3 — so yes, 315 divides by 7. It's 7 times 45. For 249: last digit 9, double it to get 18. The rest is 24, and 24 minus 18 is 6. And 6 doesn't divide by 7 — so no, 249 does not divide by 7. That's it — you now carry a seven-detector in your head.

Part E — Fractions tricks

■ Say: "Number 69" — Same Bottom → Add the Tops

Quick one — what's $\frac{3}{8}$ plus $\frac{2}{8}$? ... If you got that in a flash, you already know today's rule. If it made you freeze, don't worry — this is the friendliest fraction rule there is. When two fractions have the same bottom number, you leave the bottom completely alone and just add — or subtract — the tops. The bottom tells you the size of the pieces; the tops tell you how many pieces. Let's do $\frac{3}{8}$ plus $\frac{2}{8}$. The bottoms match — both are 8 — so the bottom stays 8. Add the tops: 3 plus 2 is 5. So $\frac{3}{8}$ plus $\frac{2}{8}$ is $\frac{5}{8}$. Done — no clever tricks needed. Your turn — pause me if you need to: what's $\frac{4}{9}$ plus $\frac{3}{9}$? ... and what's $\frac{6}{11}$ minus $\frac{2}{11}$? ... Have a proper go before I tell you... Let's check together. $\frac{4}{9}$ plus $\frac{3}{9}$: the bottoms match, so keep the 9. Add the tops: 4 plus 3 is 7. The answer is $\frac{7}{9}$. Now $\frac{6}{11}$ minus $\frac{2}{11}$: bottoms match, keep the 11. Subtract the tops: 6 minus 2 is 4. The answer is $\frac{4}{11}$. Same bottom, sort the tops — that's the whole rule. Everything else in fractions builds on this one, so you're off to a flying start.

■ Say: "Number 70" — Magic Fraction Facts

Quick fire — what's $\frac{1}{4}$ as a percentage? ... If you said 25 percent before I finished asking, you've already banked this one. If not, brilliant — today is about collecting a handful of magic facts. A few fraction facts, learned by heart, make almost every fraction, decimal and percentage question instant. Here are the stars: $\frac{1}{2}$ is 0.5, which is 50 percent. $\frac{1}{4}$ is 0.25, which is 25 percent. $\frac{3}{4}$ is 0.75, which is 75 percent. $\frac{1}{5}$ is 0.2, which is 20 percent. $\frac{1}{10}$ is 0.1, which is 10 percent. So for our opener: $\frac{1}{4}$ is 0.25 — and 0.25 means 25 out of 100, which is 25 percent. No calculating, just knowing. Your turn — pause me if you need to: what's $\frac{3}{4}$ as a percentage? ... and what's $\frac{1}{5}$ as a decimal? ... Have a proper think before I tell you... Let's check. $\frac{3}{4}$ is 0.75, so as a percentage it's 75 percent — three lots of 25. And $\frac{1}{5}$ is 0.2 as a decimal, which is 20 percent. Keep saying these facts until they pop out without thinking — then you'll barely have to calculate at all. That's a trick for life, that one.

■ Say: "Number 71" — Fraction of a Number

Quick one — what's $\frac{3}{4}$ of 20? ... If you nailed that in under ten seconds, you're already flying. If not, no bother — this trick works for any fraction of any number. To find a fraction of a number, divide by the bottom, then multiply by the top. Always divide first — it keeps the numbers nice and small. Let's do $\frac{3}{4}$ of 20. The bottom is 4, so divide: 20 divided by 4 is 5. The top is 3, so multiply: 5 times 3 is 15. So $\frac{3}{4}$ of 20 is 15. Divide by the bottom, times by the top — every single time. Your turn — pause me if you need to: what's $\frac{2}{5}$ of 30? ... and what's $\frac{3}{8}$ of 40? ... Have a proper go before I tell you... Right, let's check together. $\frac{2}{5}$ of 30: divide by the bottom — 30 divided by 5 is 6. Multiply by the top — 6 times 2 is 12. So $\frac{2}{5}$ of 30 is 12. Now $\frac{3}{8}$ of 40: divide by the bottom — 40 divided by 8 is 5. Multiply by the top — 5 times 3 is 15. So $\frac{3}{8}$ of 40 is 15. See? Bottom first, then top — you've got this one sorted.

■ Say: "Number 72" — Simplify by Halving

Quick one — can you write $\frac{12}{16}$ as a simpler fraction? ... If you got there in under ten seconds, you clearly know your halves. If not, perfect timing — here's the easiest simplifying trick there is. If the top and the bottom of a fraction are both even, halve them both. Then look again — if they're still both even, halve again. Keep going until you can't any more. Let's do $\frac{12}{16}$. Both even? Yes. Halve both: 12 becomes 6, and 16 becomes 8, giving $\frac{6}{8}$. Both still even, so halve again: 6 becomes 3, and 8 becomes 4, giving $\frac{3}{4}$. Now 3 is odd, so we stop. $\frac{12}{16}$ fully simplified is $\frac{3}{4}$. Your turn — pause me if you need to: simplify $\frac{8}{20}$... and simplify $\frac{24}{32}$... Have a proper go before I tell you... Let's check together. $\frac{8}{20}$: both even, so halve both to get $\frac{4}{10}$. Both still even — halve again to get $\frac{2}{5}$. The 5 is odd, so we're done: $\frac{8}{20}$ is $\frac{2}{5}$. Now $\frac{24}{32}$: halve both to get $\frac{12}{16}$... halve again to get $\frac{6}{8}$... and once more to get $\frac{3}{4}$. So $\frac{24}{32}$ is $\frac{3}{4}$. Halving first, every time — see, you've got this one.

■ Say: "Number 73" — Slash the Zeros

Quick one — can you make 30 over 50 simpler? ... If you saw the answer in a flash, you already know this trick. If not, no worries at all — it takes about a minute to learn. Here's the secret: if the top and the bottom of a fraction both end in 0, you can slash a zero off each one. That's really just dividing the top and bottom by 10, so the fraction stays worth exactly the same. Let's do 30 over 50. The top ends in 0, the bottom ends in 0 — slash them both off. 30 becomes 3, and 50 becomes 5. So 30 over 50 is simply 3 over 5. Done in one stroke of the pencil. Your turn — pause me if you need to: simplify 40 over 90 ... and simplify 500 over 800 ... Have a proper go before I tell you. Ready? 40 over 90: both end in 0, so slash the zeros — that's 4 over 9. And 500 over 800: slash one zero from each — 50 over 80. They still both end in 0, so slash again — 5 over 8. See? You've got this one — go slash some zeros in the practice.

■ Say: "Number 74" — Compare to One Half

Quick challenge — is 5 over 8 more than a half, or less than a half? ... If you knew that straight away, brilliant — you already own this trick. If you had to guess, stick with me for one minute. Here's the idea: a fraction is exactly a half when the top number is exactly half the bottom number. So just ask — what's half of the bottom? If the top is bigger than that, the fraction is more than a half. If it's smaller, it's less. Let's check 5 over 8. Half of 8 is 4. Is 5 more than 4? Yes — so 5 over 8 is more than a half. No sums, just a quick look. Your turn — pause me if you need to: is 3 over 10 more or less than a half? ... and is 7 over 12 more or less than a half? ... Have a proper think before I tell you. Right — 3 over 10: half of 10 is 5, and 3 is less than 5, so 3 over 10 is less than a half. And 7 over 12: half of 12 is 6, and 7 is more than 6, so 7 over 12 is more than a half. That's it — you can now size up a fraction at a glance.

■ Say: "Number 75" — Cross-Multiply to Compare

Quick one — which is bigger, $\frac{2}{3}$ or $\frac{5}{8}$? ... If you cracked that in under ten seconds, hats off — you know this already. If not, you're about to learn a proper ninja move. To compare two fractions, cross-multiply: each top times the OTHER fraction's bottom. Whichever side makes the bigger product wins. It works because you're secretly making a common denominator without writing it down. Let's settle $\frac{2}{3}$ against $\frac{5}{8}$. Take 2 times 8 — that's 16. Now 5 times 3 — that's 15. 16 is bigger, and the 16 came from the $\frac{2}{3}$ side — so $\frac{2}{3}$ is the bigger fraction. Your turn — pause me if you need to: which is bigger, $\frac{3}{4}$ or $\frac{5}{7}$? ... and which is bigger, $\frac{2}{5}$ or $\frac{3}{7}$? ... Have a proper go before I spill the answers. Here we go. $\frac{3}{4}$ against $\frac{5}{7}$: 3 times 7 is 21, and 5 times 4 is 20. 21 wins, and it came from $\frac{3}{4}$ — so $\frac{3}{4}$ is bigger. Next, $\frac{2}{5}$ against $\frac{3}{7}$: 2 times 7 is 14, and 3 times 5 is 15. 15 wins, so $\frac{3}{7}$ is bigger. See? You've got this one — go smash the practice.

■ Say: "Number 76" — Cancel Before Multiplying

Quick one — what's $\frac{3}{4}$ times $\frac{8}{9}$? ... If that felt easy, you already know today's trick. If it felt scary, don't worry — this is the trick that makes it easy. When you multiply fractions, look for shared factors and slash them out FIRST, before you multiply. You can cancel any top with any bottom — diagonally or in the same fraction — but never top with top. Let's do $\frac{3}{4}$ times $\frac{8}{9}$. The 3 on top and the 9 on the bottom share a 3 — cancel them down to 1 and 3. The 8 on top and the 4 on the bottom share a 4 — cancel them down to 2 and 1. Now it's just 1 over 1 times 2 over 3, which is $\frac{2}{3}$. Tiny numbers, no mess. Your turn — pause me if you need to: what's $\frac{2}{3}$ times $\frac{9}{10}$? ... and what's $\frac{5}{6}$ times $\frac{3}{10}$? ... Have a proper go first. Right — $\frac{2}{3}$ times $\frac{9}{10}$: the 2 and the 10 cancel to 1 and 5, the 9 and the 3 cancel to 3 and 1, leaving $\frac{3}{5}$. And $\frac{5}{6}$ times $\frac{3}{10}$: the 5 and the 10 cancel to 1 and 2, the 3 and the 6 cancel to 1 and 2, leaving 1 over 2 times 1 over 2 — that's $\frac{1}{4}$. That's it — cancel first, and multiplying fractions becomes a doddle.

■ Say: "Number 77" — Keep, Change, Flip

Quick challenge — what's a half divided by a quarter? ... If you said 2 without blinking, you already know this famous trick. If dividing fractions makes your brain wobble, perfect — this fixes it forever. The trick is called Keep, Change, Flip. KEEP the first fraction exactly as it is. CHANGE the divide sign into a times sign. FLIP the second fraction upside down. Then just multiply. Let's do 1 over 2 divided by 1 over 4. Keep the 1 over 2. Change divide to times. Flip 1 over 4 into 4 over 1. Now multiply: 1 over 2 times 4 over 1 is 4 over 2, and 4 over 2 is 2. So a half divided by a quarter is 2 — because two quarters fit inside a half. Your turn — pause me if you need to: what's 3 over 4 divided by 1 over 8? ... and what's 1 over 3 divided by 1 over 6? ... Have a proper go before I tell you. Ready? 3 over 4 divided by 1 over 8: keep 3 over 4, change to times, flip to 8 over 1. That's 24 over 4, which is 6. And 1 over 3 divided by 1 over 6: keep 1 over 3, times, flip to 6 over 1 — that's 6 over 3, which is 2. See? Keep, change, flip — you now know a trick most grown-ups have forgotten.

■ Say: "Number 78" — Butterfly Method

Quick one — what's 2 over 3 plus 1 over 4? ... If you got 11 over 12 in under ten seconds, you're flying — you know this one. If not, let me show you the butterfly. When two fractions have different bottoms, draw a butterfly: multiply across the wings to get the tops, and multiply the two bottoms for the new bottom. No hunting for a common denominator — the butterfly does it for you. Let's do 2 over 3 plus 1 over 4. First wing: 2 times 4 is 8. Second wing: 1 times 3 is 3. Add those for the top: 8 plus 3 is 11. Now the bottoms: 3 times 4 is 12. So 2 over 3 plus 1 over 4 is 11 over 12. Your turn — pause me if you need to: what's 1 over 2 plus 1 over 5? ... and what's 3 over 5 plus 1 over 4? ... Have a proper go before I tell you. Here we go. 1 over 2 plus 1 over 5: wings first — 1 times 5 is 5, and 1 times 2 is 2. Top: 5 plus 2 is 7. Bottom: 2 times 5 is 10. Answer: 7 over 10. And 3 over 5 plus 1 over 4: 3 times 4 is 12, 1 times 5 is 5, top is 12 plus 5 — 17. Bottom: 5 times 4 is 20. Answer: 17 over 20. That's it — flap those wings and go smash the practice.

■ Say: "Number 79" — Spot the Easy Common Bottom

Quick one — what's 1 over 4 plus 3 over 8? ... If you spotted the shortcut and got 5 over 8, you're already ahead. If not, here's the little check that saves loads of work. Before you reach for any fancy method, look at the two bottoms and ask: is one a multiple of the other? If it is, you only need to change ONE fraction, and the sum becomes easy. Let's do 1 over 4 plus 3 over 8. Is 8 a multiple of 4? Yes — 4 times 2 is 8. So change 1 over 4 into eighths: multiply top and bottom by 2, and 1 over 4 becomes 2 over 8. Now the bottoms match: 2 over 8 plus 3 over 8 is 5 over 8. Done. Your turn — pause me if you need to: what's 1 over 3 plus 2 over 9? ... and what's 1 over 2 plus 3 over 8? ... Have a proper go before I tell you. Right then. 1 over 3 plus 2 over 9: 9 is a multiple of 3, so 1 over 3 becomes 3 over 9. Then 3 over 9 plus 2 over 9 is 5 over 9. And 1 over 2 plus 3 over 8: 8 is a multiple of 2, so 1 over 2 becomes 4 over 8. Then 4 over 8 plus 3 over 8 is 7 over 8. See? Always hunt for the easy common bottom first — the butterfly is just the back-up plan.

■ Say: "Number 80" — Fraction → Percent

Quick challenge — what's 3 over 5 as a percentage? ... If you said 60 percent before I finished asking, you already own this trick. If not, one minute from now you will. Per cent just means out of 100. So to turn a fraction into a percentage, make the bottom into 100 — multiply the top and the bottom by the same number until the bottom hits 100. Whatever lands on top is your percentage. Let's do 3 over 5. What turns 5 into 100? Multiply by 20. So multiply the top by 20 as well: 3 times 20 is 60. That gives 60 over 100 — which is exactly 60 percent. So 3 over 5 is 60 percent. Your turn — pause me if you need to: what's 7 over 20 as a percentage? ... and what's 9 over 25 as a percentage? ... Have a proper go before I tell you. Ready? 7 over 20: to get from 20 to 100, multiply by 5. So 7 times 5 is 35 — that's 35 over 100, which is 35 percent. And 9 over 25: 25 times 4 is 100, so 9 times 4 is 36 — that's 36 over 100, which is 36 percent. That's it — you can now turn fractions into percentages faster than a calculator comes out of the drawer.

Part F — Percentages tricks

■ Say: "Number 81" — 10% — Shift the Decimal

Quick one — what's 10 percent of 360? ... If you said 36 in a heartbeat, you already know the most important percentage trick there is. If not, brilliant timing — because this one unlocks nearly everything else. 10 percent means 1 out of every 10, so finding 10 percent is just dividing by 10. And the quickest way to divide by 10 is to move the decimal point one place to the left. Let's do 10 percent of 360. Picture the decimal point sitting after the 360. Slide it one place left and you get 36. So 10 percent of 360 is 36. That's the whole trick — one little slide. Your turn — pause me if you need to: what's 10 percent of 250? ... and what's 10 percent of 42? ... Have a proper go before I tell you. Right — 10 percent of 250: slide the decimal point one place left and 250 becomes 25. So 10 percent of 250 is 25. And 10 percent of 42: slide the point one place left and 42 becomes 4.2. So 10 percent of 42 is 4.2 — the point just tucks in between the 4 and the 2. See? You've got this one — and almost every other percentage is built from it, so go smash the practice.

■ Say: "Number 82" — 1% — Shift TWICE

Quick one — what's 1% of 300? ... If that popped out in a few seconds, you already know this trick. If not, no bother at all — you'll have it in a minute. Here's the idea. 1% means 1 out of every 100, so finding 1% is just dividing by 100. And the speedy way to divide by 100 is to move the decimal point two places to the left. That's the whole trick. Let's do 1% of 300. Slide the decimal point two places left, and 300 becomes 3. So 1% of 300 is 3. Your turn — pause me if you need to: what's 1% of 500? ... and what's 1% of 62? ... Have a proper go before I tell you. ... Ready? 1% of 500: move the decimal point two places left, and 500 becomes 5. So 1% of 500 is 5. Now 1% of 62: two hops left turns 62 into 0.62. So 1% of 62 is 0.62. And here's the best bit — once you can find 1%, you can build any percentage at all. See? Two little hops and you're there — go smash the practice.

■ Say: "Number 83" — % ↔ Fraction Magic

Here's a quick one — what's 25% of 80? ... Got it in under ten seconds? Then you already think like this trick. If not, stick with me — it's a lovely one. The secret is that some percentages are really fractions in disguise. 50% is a half, 25% is a quarter, 75% is three quarters, 20% is a fifth, and 10% is a tenth. Swap the percentage for its fraction and the sum turns easy. So, 25% of 80: 25% is a quarter, and a quarter of 80 is 80 divided by 4, which is 20. So 25% of 80 is 20 — no long working, just one divide. Your turn — pause me if you need to: what's 50% of 90? ... and what's 20% of 35? ... Have a proper think before I tell you. ... Right, let's check. 50% is a half, and half of 90 is 45. So 50% of 90 is 45. Now 20% is a fifth, and 35 divided by 5 is 7. So 20% of 35 is 7. That's it — you now know a shortcut plenty of grown-ups still reach for a calculator to do.

■ Say: "Number 84" — Build from 10, 5 and 1

Try this — what's 15% of 60? ... If you rattled that off, brilliant, you've already got the idea. If it felt tricky, don't worry — this trick makes it easy. You can build any percentage out of little blocks: 10%, 5% and 1%. 10% is just divide by 10, 5% is half of your 10%, and 1% is divide by 100. Stack the blocks until you reach the percentage you want. So, 15% of 60. 10% of 60 is 6. 5% is half of that, which is 3. Add them: 6 plus 3 is 9. So 15% of 60 is 9. Your turn — pause me if you need to: what's 15% of 40? ... and what's 12% of 300? ... Have a proper go first. ... Let's check together. 15% of 40: 10% of 40 is 4, 5% is half of that, 2, and 4 plus 2 is 6. So 15% of 40 is 6. Now 12% of 300: 10% of 300 is 30, 1% is 3, so 2% is 6, and 30 plus 6 is 36. So 12% of 300 is 36. See? Little blocks, big answers — you've got this one.

■ Say: "Number 85" — Halve and Double

Quick challenge — what's 20% of 45? ... If that came out in under ten seconds, you're already halfway to mastering this. If not, no stress — let's learn it right now. The trick is to find 10% first, by dividing by 10. Then two quick moves unlock loads more: halve the 10% to get 5%, or double it to get 20%. Most everyday percentages are just a halve or a double away. So, 20% of 45. 10% of 45 is 4.5. Double it: 4.5 plus 4.5 is 9. So 20% of 45 is 9. Your turn — pause me if you need to: what's 5% of 60? ... and what's 20% of 35? ... Have a proper go before I spill the answers. ... Okay, checking time. 5% of 60: 10% of 60 is 6, and half of 6 is 3. So 5% of 60 is 3. Now 20% of 35: 10% of 35 is 3.5, and double 3.5 is 7. So 20% of 35 is 7. See? Find 10%, then halve or double — that's most percentages sorted for life. Go smash the practice.

■ Say: "Number 86" — Swap the Numbers

Here's a sneaky one — what's 4% of 50? ... If you got 2 straight away, you might already know this magic. If it looked horrible, perfect — watch this. There's a magical fact about percentages: a certain percent of one number is always the same as swapping the numbers round. So 4% of 50 is exactly the same as 50% of 4. When a percentage looks awkward, flip it and it often turns friendly. Let's do it. 4% of 50 — swap it — becomes 50% of 4. And 50% is just a half, so half of 4 is 2. So 4% of 50 is 2. Done in one breath. Your turn — pause me if you need to: what's 8% of 25? ... and what's 6% of 50? ... Have a proper go before I tell you. ... Ready? 8% of 25 — swap it — is 25% of 8. That's a quarter of 8, which is 2. So 8% of 25 is 2. Now 6% of 50 — swap it — is 50% of 6, which is half of 6, so 3. So 6% of 50 is 3. See? You now know a trick most grown-ups have never even heard of.

■ Say: "Number 87" — "Is over Of"

Quick one — you score 30 out of 40 on a test. What percent is that? ... If you said 75% in a flash, you've already got this. If not, no worries — here comes the trick. To turn a score into a percentage, take the part, divide it by the whole, then multiply by 100. Remember it as is over of: what percent IS 30 OF 40 means 30 over 40. Let's walk it through. 30 divided by 40 is 0.75. Then 0.75 times 100 is 75. So 30 out of 40 is 75%. Your turn — pause me if you need to: what percent is 18 out of 20? ... and what percent is 15 out of 60? ... Have a proper go before I tell you. ... Right, together now. 18 out of 20: 18 divided by 20 is 0.9, and 0.9 times 100 is 90. So 18 out of 20 is 90% — a cracking score. Now 15 out of 60: 15 divided by 60 is 0.25, and 0.25 times 100 is 25. So 15 out of 60 is 25%. That's it — every test score you ever get, you can turn into a percent in seconds.

■ Say: "Number 88" — Multiplier Method (+% or -%)

Try this — take £80 and add 25%. What do you get? ... Under ten seconds? Then you're already using this trick without knowing it. If not, let's learn it right now. To add a percentage in one hit, multiply by 1 plus the percentage as a decimal — so plus 25% means times 1.25, and plus 10% means times 1.1. To take a percentage off, multiply by 1 minus the decimal — minus 25% means times 0.75, minus 10% means times 0.9. One multiply, no extra adding step. So, £80 plus 25%: that's 80 times 1.25. 80 times 1 is 80, and a quarter of 80 is 20, so 80 plus 20 is 100. £80 plus 25% is £100. Your turn — pause me if you need to: what's £60 plus 50%? ... and what's £40 minus 10%? ... Have a proper go first. ... Let's check. £60 plus 50% is 60 times 1.5: 60 plus half of 60, which is 30, makes 90. So £90. Now £40 minus 10% is 40 times 0.9: 40 minus 4 is 36. So £36. See? One multiplier does the whole job — go smash the practice.

■ Say: "Number 89" — Chain Two %s with Multipliers

Here's a brain-teaser — prices go up 20%, then down 20%. Are you back where you started? ... If you smelled a trick and said no, you're sharp — most grown-ups get this wrong. If you said yes, brilliant, because you're about to find out why it isn't. The trick: turn each change into a multiplier, then multiply the multipliers together. Up 20% is times 1.2, down 20% is times 0.8. Let's see. 1.2 times 0.8 is 0.96. Times 0.96 means you end up with 96% of what you started with — so it's actually a 4% loss overall, not back to the start. Sneaky! Your turn — pause me if you need to: up 10% then up 10% again — what's the total change? ... and up 50% then down 50%? ... Have a proper think first. ... Let's check. Up 10% twice is 1.1 times 1.1, which is 1.21 — a 21% rise, a bit more than 20. And up 50% then down 50% is 1.5 times 0.5, which is 0.75 — a 25% loss. See? You can now out-think a sale sign — that's proper maths power.

■ Say: "Number 90" — Reverse Percentages

Quick one — a game costs £45 in a half-price sale. What was the full price? ... If you said £90 straight off, you've already got the knack. If not, no worries — this trick works backwards through any sale. Here's how: work out what percentage is LEFT after the discount, turn it into a decimal, and divide the sale price by it. That undoes the change and gives you the original. So, the £45 game: half price means 50% is left, and 50% as a decimal is 0.5. Then 45 divided by 0.5 is 90. So the full price was £90. Your turn — pause me if you need to: a jacket is £60 after 25% off — what was the original price? ... and a book is £27 after 10% off? ... Have a proper go before I tell you. ... Let's check together. 25% off means 75% is left, which is 0.75. And 60 divided by 0.75 is 80 — the jacket was £80. Now the book: 10% off means 90% left, which is 0.9. And 27 divided by 0.9 is 30 — it was £30. See? You can rewind any sale — that's a trick shopkeepers hope you never learn.

■ Say: "Number 91" — Tip Combos

Quick one — you're out for pizza and the bill is £40. What's a 15 percent tip? ... If that popped out in under ten seconds, you're already a tipping champion. If not, stick with me — this one is genuinely useful for life. Here's the secret: every tip is built from 10 percent. 15 percent is 10 percent plus half of it again. 20 percent is just 10 percent doubled. Find 10 percent first and the rest falls out. So, that £40 bill. 10 percent of 40 is 4 — that's £4. Half of £4 is £2. £4 plus £2 is £6. A 15 percent tip on £40 is £6. Your turn — pause me if you need to: what's a 20 percent tip on a £35 bill? ... and what's a 15 percent tip on a £60 bill? ... Have a proper go before I tell you. Ready? The £35 bill first: 10 percent of 35 is £3.50. Double it for 20 percent: £7. Now the £60 bill: 10 percent of 60 is £6. Half of £6 is £3. £6 plus £3 is £9 — that's your 15 percent. See? Next time you're out for dinner, you'll have the tip sorted before the grown-ups find their phones.

■ Say: "Number 92" — Use a Fraction for Tidy %s

Here's a quick one — what's 75 percent of 80? ... If you got that in under ten seconds, you already think like this trick. If not, no worries at all — you're about to learn a lovely shortcut. Some percentages are much faster as fractions. 25 percent is a quarter, 50 percent is a half, 75 percent is three quarters, and 60 percent is three fifths. Swap the percentage for its fraction, and a scary sum turns friendly. Let's do 75 percent of 80. 75 percent is three quarters. 80 divided by 4 is 20. 20 times 3 is 60. So 75 percent of 80 is 60. Your turn — pause me if you need to: what's 25 percent of 64? ... and what's 60 percent of 45? ... Have a proper go before I tell you. Right, let's check together. 25 percent is a quarter, so 64 divided by 4 is 16 — that's it, 16. And 60 percent is three fifths: 45 divided by 5 is 9, and 9 times 3 is 27. So 60 percent of 45 is 27. That's it — you now know a trick most grown-ups don't, so go smash the practice.

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